Generative Modeling for Robust Speaker Diarization

PhD Defense

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Outline

1. Overview of speaker diarization
2. Paradigms of speaker diarization
3. Neural FCASA: joint modeling of diarization and separation
4. Our contributions: Beta FCASA and Heavy-tailed FCASA
5. Conclusion

What is Speaker Diarization?

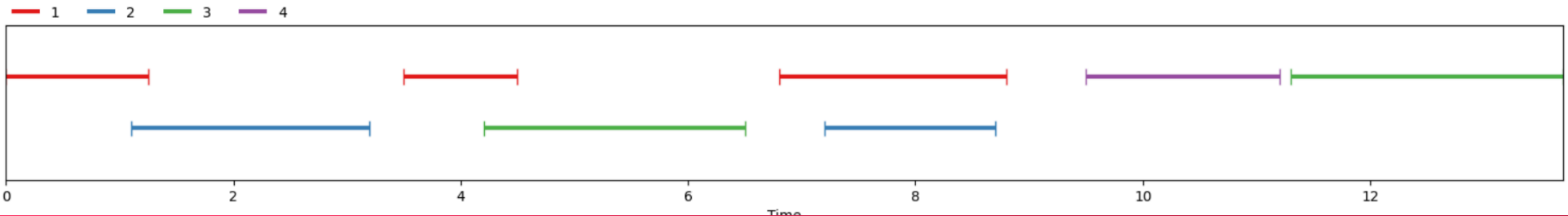
Objective: Decide who speaks when

- Who: *Identify* the active speakers.
- When: Find the *boundary* of their speech/voice activity.
- Application: Meeting transcription.

Data

- Input: Mono-channel/Multi-channel audio
- Output: Rich Transcription Time Marked (RTTM) file

```
file type, file name, channel ID, begin time, duration, speaker ID.
SPEAKER,   ES2005a,   1,          0.00,       1.25,     1,
SPEAKER,   ES2005a,   1,          1.10,       2.10,     2,
SPEAKER,   ES2005a,   1,          3.50,       1.00,     1,
```



Challenges of speaker diarization

Speaker diarization is a special case of pattern recognition problem.

Difference versus Identity

- The core problem in pattern recognition in general.
- Distinguish two types of variability:
  - **Intra-class variability**: differences that should be ignored, *style*.
  - **Inter-class variability**: differences that should be emphasized, *content*.
- The two types of variability are **entangled**.



Challenges of speaker diarization

Many factors contribute to the difficulty of speaker diarization problem.

Speech variability in general

- phone, prosody, emotion
- timbre, accent
- They are entangled.

Speech variability in meeting scenario

- noise, reverberation.
- overlap speech, unknown number of speakers.

Data scarcity

- Limited labeled data, e.g., AMI corpus (~100 h).
- Low effective information, e.g., 3 mins of audio $\approx$ 10 million of characters $\approx$ 30MB.

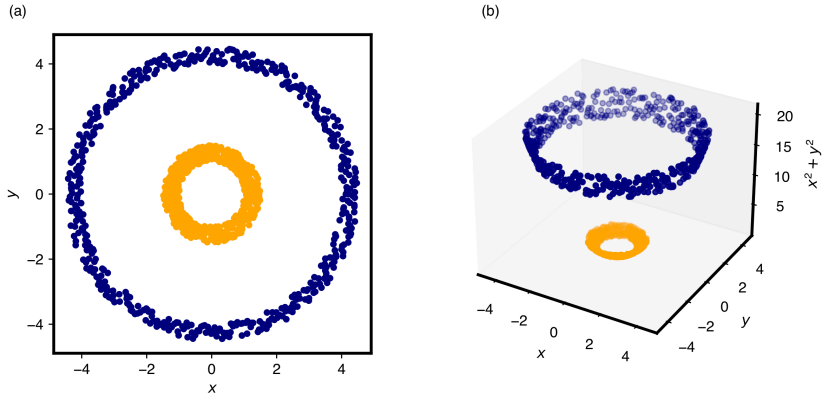
How to address these challenges

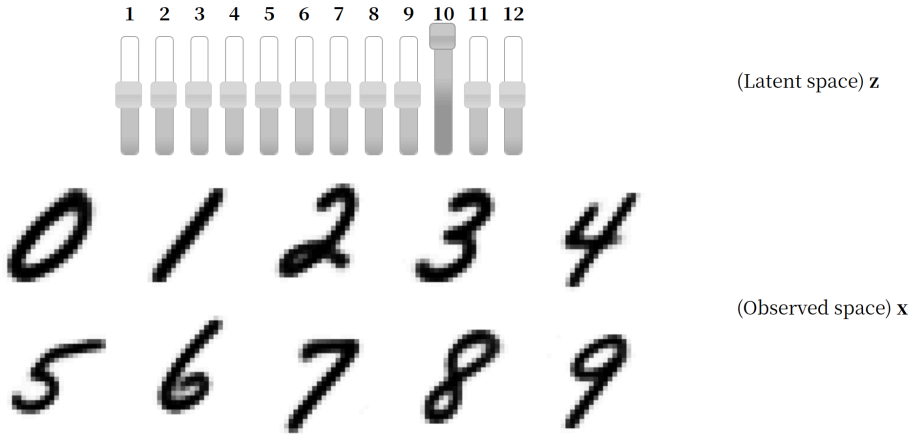
Entangled variabilities -> representation learning

- Minimize intra-speaker variability and maximize inter-speaker variability.
- Representations are more separable.

Data scarcity -> generative modeling

- Incorporate prior knowledge.
- Compress information.





Evaluation metrics

Diarization Error Rate (DER):

Evalutes three types of errors:

- **Missed** (Miss): a frame with speaker -> silence.
- **False alarm** (FA): a silent frame -> some speaker.
- **Speaker confusion** (Conf.): a frame with speaker A -> speaker B.

$$\mathrm{DER} = \frac{\mathrm{FA} + \mathrm{Miss} + \mathrm{Conf.}}{\mathrm{Total\ annotated\ duration}}$$

Jaccard Error Rate (JER)

Evaluates the error rate speaker-wise:

$$\mathrm{JER} = \frac{1}{N_{ref}} \sum_{i}^{N_{ref}} \frac{\mathrm{FA_i} + \mathrm{Miss_i}}{\mathrm{TOTAL_i}}$$

Speaker diarization paradigms

Paradigms based on the modeling approach

- Data-driven:
  - Large data for end-to-end training (e.g. EEND [Fujita et al., 2019])
  - Large parameter with LLM backbone (e.g. VibeVoice-ASR [Peng et al., 2026])
- Model-driven/Hybrid: Exploit prior knowledge with DSP models.

Paradigms based on the modeling objective

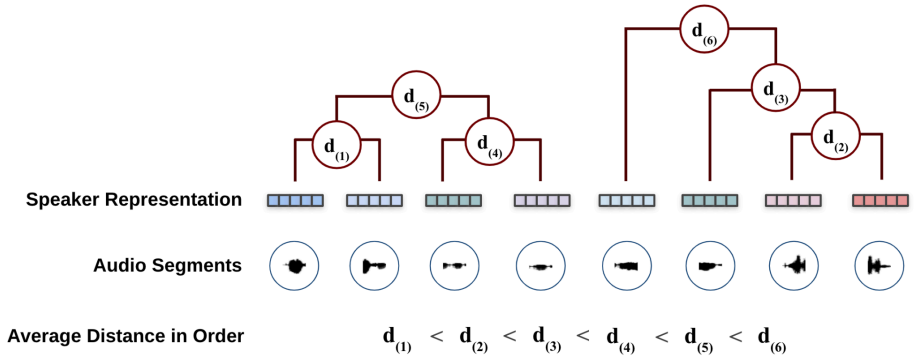
- Segmentation-clustering
- Multi-label classification
- Text-generation

Our contributions fall in the model-driven and multi-label classification paradigm.

Segmentation-clustering paradigm

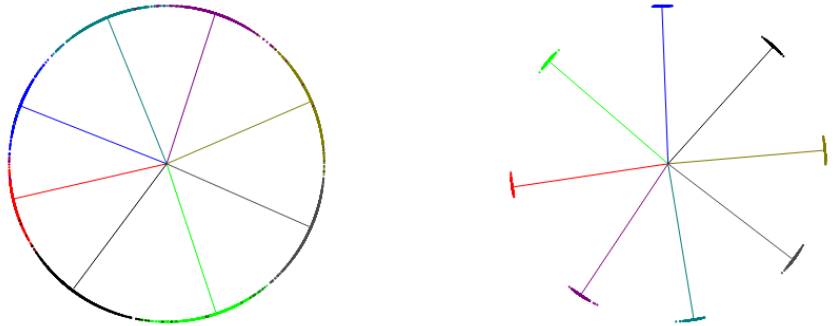
Pipeline

1. Cut the audio into short segments.
2. Extract speaker representations from the segments.
3. Cluster the speaker representations based on similarity scores.



Core components

- Speaker embedding extractor
  - Representation learning with discriminative loss.
  - Trained on single speaker recordings only.
- Scoring method: cosine similarity v.s. PLDA.
- Clustering algorithm: AHC v.s. VB-HMM.

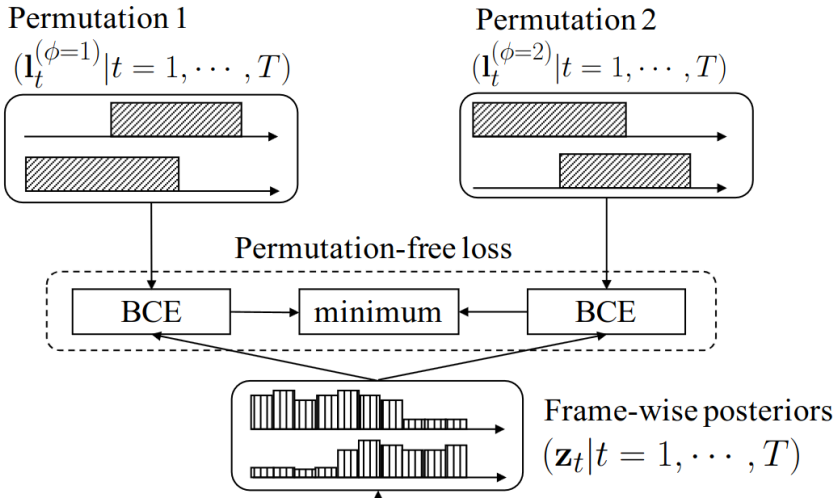


(a) Softmax (b) ArcFace

Multi-label classification paradigm

End-to-end neural diarization (EEND) [Fujita et al., 2019]

- Per-frame multi-label classification formulation.
- Permutation-invariant training (PIT) handles the label ambiguity.
- Model the speaker- & time- variability.
- Large synthetic data.



Following works

- Stronger time-modeling backbone: EEND-EDA [Horiguchi et al., 2022a].
- Leverage multi-channel information: Multi-channel EEND [Horiguchi et al., 2022b].
- Joint source separation and diarization: EEND-SS [Maiti et al., 2023].

However, these models are still problematic: data-hungry, synthetic v.s. real gap.

Model-driven diarization?

Q: Can we incorporate prior knowledge into diarization model?
A: We have little knowledge on the **speaker identity** or the **conversational dynamics** 😕.

Q: Can other task help?
A: Yes! For example the source separation (SS) task.

- Intuitively, perfect separation makes the diarization simply VAD.
- Experimentally, EEND-SS proves the effectiveness.

Q: Do we have prior knowledge on the auxiliary task?
A: Yes! SS has been well studied during many years and there are many DSP models for it.

So we need to investigate model-driven source separation first!

Model-driven separation

A common **generative model** for multichannel SS [Vincent 2011]. Consider the representation in TF domain with f and t as the frequency and time indices.

Denote λ_{nft} the **PSD** of speaker n, the signal s_{nft} is generated from the **Gaussian variance modeling**:

$$s_{nft} \sim \mathcal{N}_{\mathbb{C}}\left(0, \lambda_{nft}\right),$$

The image signal $\mathbf{x}_{nft}$ is generated from the source signal s_{nft} modulated by the **SCM** $\mathbf{R}_{nf}$:

$$\mathbf{x}_{nft} \sim \mathcal{N}_{\mathbb{C}}\left(0, \lambda_{nft}\mathbf{R}_{nf}\right).$$

The mixture signal $\mathbf{x}_{ft}$ is the sum of the image signals of all speakers:

$$\mathbf{x}_{ft} = \sum_{n=1}^{N} \mathbf{x}_{nft} \sim \mathcal{N}_{\mathbb{C}}\left(0, \mathbf{Y}_{ft}\right).$$

where $\mathbf{Y}_{nft} = \lambda_{nft}\mathbf{R}_{nf}$ and $\mathbf{Y}_{ft} = \sum_{n=1}^{N} \mathbf{Y}_{nft}$.

Source separation with hybrid models

Neural Full-rank Covariance Analysis (Neural FCA) [Bando et al., 2023]

A **deep generative model** for the source PSD, with a deep spectral feature $\mathbf{z}_{nt}$ and a neural network $g_{\theta,f}(\cdot)$:

$$\begin{aligned}\mathbf{z}_{nt} &\sim \mathcal{N}(\mathbf{0}, \mathbf{I}),\\ \lambda_{nft} &= g_{\theta,f}(\mathbf{z}_{nt}),\\ \mathbf{x}_{ft} &\sim \mathcal{N}_{\mathbb{C}}\left(0, \mathbf{Y}_{ft}\right).\end{aligned}$$

Hybrid inference

- Infer g with variational autoencoder (VAE) [Kingma et al., 2014a].
- Infer λ_{nft} and $\mathbf{R}_{nf}$ with iterative source steering method (ISS) [Scheibler et al., 2020].

Blind source separation: completely unsupervised, no need of clean separated signals.

Joint separation and diarization

Neural FCA with Speaker Activity (Neural FCASA) [Bando et al., 2024]

A joint separation and diarization framework that incorporates speaker activity mask $u_{nt} \in \{0, 1\}$:

$$\mathbf{x}_{ft} \sim \mathcal{N}_{\mathbb{C}}\left(0, \mathbf{Y}_{ft}\right).$$

where $\mathbf{Y}_{ft} = \sum_{n=1}^{N} u_{nt} \lambda_{nft} \mathbf{R}_{nf}$.

- It only needs speaker annotated mixture signals for training.
- It is data-efficient(90h) and parameter-efficient(~20M).

From a separation perspective: a separation model with diarization annotations as weak supervision.

From a diarization perspective: a diarization model incorporating prior knowledge from SS.

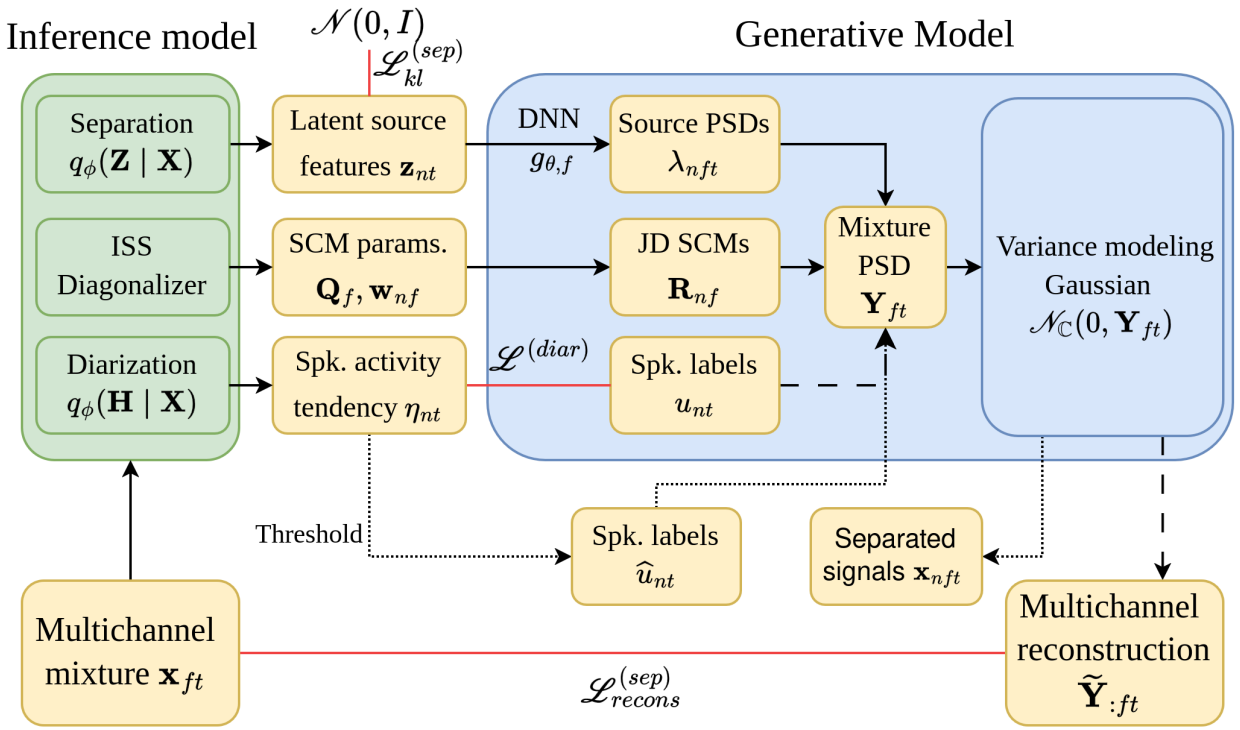
Objective of neural FCASA

- Diarization: $p(\mathbf{U} \mid \mathbf{X})$
- Separation: $p(\mathbf{Z} \mid \mathbf{X})$

They are intractable -> variational inference.

Surrogate models:

- $q_\phi(\mathbf{Z} \mid \mathbf{X}) \triangleq \prod_{n,t,d=1}^{N,T,D} \mathcal{N}(z_{ntd}\ ;\ \mu_{\phi,ntd}, \sigma^2_{\phi,ntd})$
- $q_\phi(\mathbf{U} \mid \mathbf{X}) \triangleq \prod_{n,t=1}^{N,T} \text{Bernoulli}(u_{nt}; \eta_{\phi,nt})$



Variational Inference (VI)

Optimize evidence lower bound (ELBO):

$$\log p_\theta(\mathbf{X}, \mathbf{U}) \geq \underbrace{\mathbb{E}_{q_\phi(\mathbf{Z}|\mathbf{X})}[\log p_\theta(\mathbf{X} \mid \mathbf{U}, \mathbf{Z})]}_{\mathcal{L}_{recons}^{(sep)}} \underbrace{-\mathrm{KL}[q_\phi(\mathbf{Z} \mid \mathbf{X}) \mid\mid p_\theta(\mathbf{Z})]}_{\mathcal{L}_{kl}^{(sep)}} + \underbrace{\log p(\mathbf{U})}_{\mathcal{L}^{(diar)}},$$

$\mathcal{L}_{recons}^{(sep)}$ and $\mathcal{L}_{kl}^{(sep)}$: ELBO [Kingma et al., 2014a] for separation model, with a teacher-forcing trick ($\mathbf{U}$).

$\mathcal{L}^{(diar)}$: The prior $p(\mathbf{U})$ is not defined in the generative model, instead, use $\log q_\phi(\mathbf{U}|\mathbf{X})$ as an approximation. Similar to semi-supervised VAE loss [Kingma et al., 2014b].

Q: Define a prior model for the speaker label $\mathbf{U}$ to enable VI for the diarization model as well?

Prior model for speaker activity

As the speaker activity mask $\mathbf{U}$ is a binary variable, we propose to model it as a Bernoulli variable with a Beta prior:

$$\eta_{nt} \sim \text{Beta}(\alpha, \beta)$$
$$u_{nt} \sim \text{Bernoulli}(\eta_{nt})$$

Intuitively, the prior model may capture prior knowledge about the speaker activity patterns, including:

1. the speaker's character, e.g., whether he/she is outgoing or shy to speak,
2. the ambiance of the conversation, e.g., whether it is encouraging or discouraging for the attendees to speak.

Combined with neural FCASA, we call it **Beta FCASA** [Mao et al., 2026a][1].

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1. Neural Multichannel Distant Speaker Diarization and Source Separation with Beta Speaker Activity Prior ↩

VI in Beta FCASA

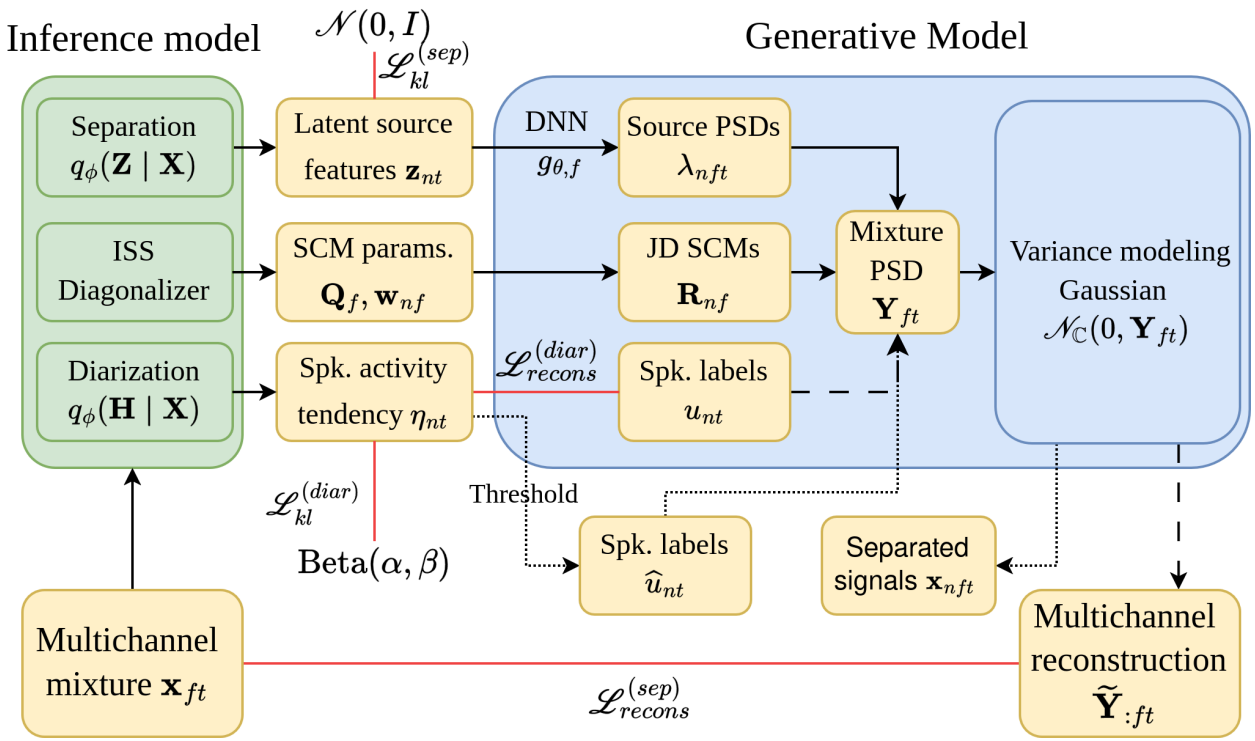
Inference model:

$$q_\phi(\mathbf{H} \mid \mathbf{X}) \triangleq \prod_{n,t=1}^{N,T} \text{Beta}\left(\eta_{nt} \mid \alpha_{\phi,nt}, \beta_{\phi,nt}\right),$$

$\mathcal{L}^{(diar)}$ becomes:

$$\underbrace{\mathbb{E}_{q_\phi(\mathbf{H})}[\log p_\theta(\mathbf{U} \mid \mathbf{H})]}_{\mathcal{L}^{(diar)}_{recons}} - \underbrace{D_{\mathrm{KL}}(q_\phi(\mathbf{H}) \| p_\theta(\mathbf{H}))}_{\mathcal{L}^{(diar)}_{kl}}$$

i.e., for the diarization likelihood, we optimize explicitly its ELBO function.



Tractablitity of the ELBO

- The separation part: standard (Gaussian) VAE ELBO, tractable (See VAE paper [Kingma et al., 2014a])
- The diarization part:
  - $\mathcal{L}_{recons}^{(diar)}$: use Beta-Bernoulli conjugacy

$$u\psi(\alpha_\phi) + (1-u)\psi(\beta_\phi) - \psi(\alpha_\phi + \beta_\phi)$$

  - $\mathcal{L}_{kl}^{(diar)}$: use the closed-form of KL divergence between two Beta distributions

$$\ln \frac{\Gamma(\alpha_\phi + \beta_\phi)\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)\Gamma(\alpha_\phi)\Gamma(\beta_\phi)} + (\alpha_\phi - \alpha)\psi(\alpha_\phi) + (\beta_\phi - \beta)\psi(\beta_\phi) - (\alpha_\phi + \beta_\phi - \alpha - \beta)\psi(\alpha_\phi + \beta_\phi)$$

  where $\psi(\cdot)$ is the digamma function and $\Gamma(\cdot)$ is the gamma function.

A subtlety of the Beta distribution

Under the canonical parametrization, the PDF of Beta distribution is:

$$\mathrm{Beta}(u;\alpha,\beta)=\frac{u^{\alpha-1}(1-u)^{\beta-1}}{B(\alpha,\beta)},\quad u\in(0,1)$$

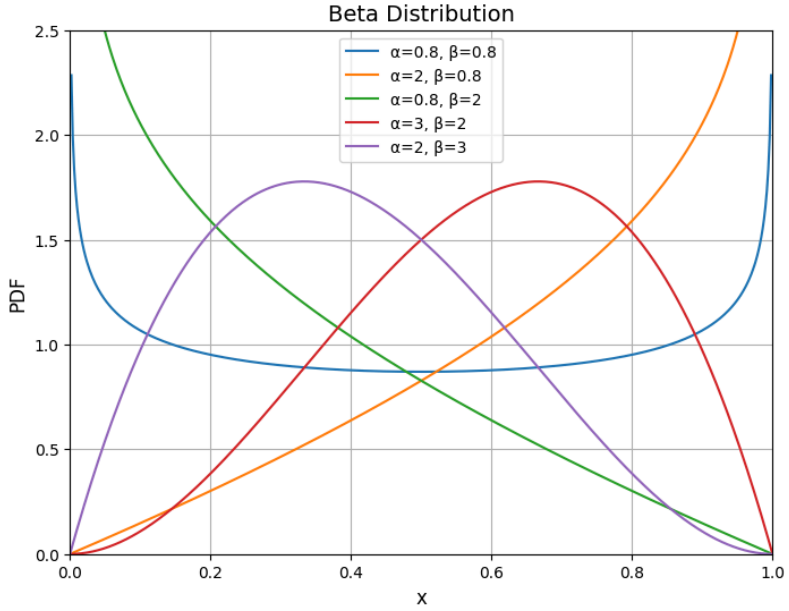
- the parameters α and β do not directly control the shape of the PDF.
- Reparametrization in terms of *mode* m and *concentration* λ by the PERT parametrization [Malcolm et al., 1959]:

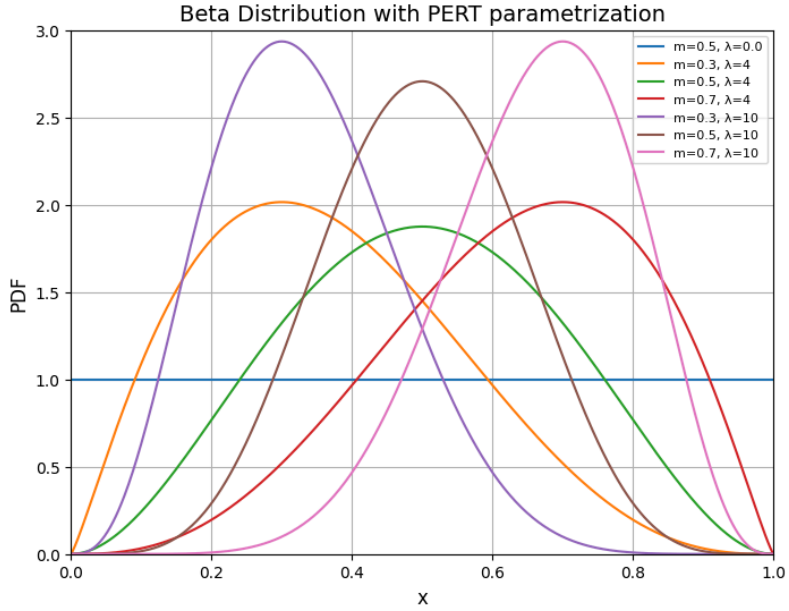
$$\begin{aligned}\mathrm{PERT}&:(\alpha,\beta)\mapsto\left(\frac{\alpha-1}{\alpha+\beta-2},\alpha+\beta-2\right),\\ \mathrm{PERT}^{-1}&:(m,\lambda)\mapsto(1+\lambda m,1+\lambda(1-m))\end{aligned}$$

This is a differentiable bijection for $\alpha,\beta>1$.

A subtlety of the Beta distribution

An illustration of the shape of Beta distribution w.r.t. its parametrization:





In short, under PERT parametrization, we can directly optimize the shape.

Experiments and results

We choose different m, λ for the prior model $p(\mathbf{H})$, and verify our methods on the AMI dataset.

- Consistently better performance than the baseline.
- λ has less impact than m.
- Efficacy on AliMeeting and CHiME-6 datasets as well.

Results on AMI dataset

m, λ	Setup	Miss (↓)	FA (↓)	Conf. (↓)	DER (↓)	JER (↓)
(baseline)	Forgiving	5.59	8.16	0.73	14.48	13.50
	Fair	8.61	6.33	0.79	15.73	23.16
	Full	10.77	6.81	1.16	18.73	27.65
	Overlap	19.40	3.80	0.90	24.11	26.53
0.3, 4	Forgiving	5.37	**4.31**	**0.43**	**10.11**	**9.73**
	Fair	8.00	**3.39**	**0.44**	**11.84**	**16.85**
	Full	9.96	**4.70**	**0.65**	15.31	**21.98**
	Overlap	17.20	**2.71**	**0.49**	20.41	**21.90**
0.5, 4	Forgiving	**4.55**	5.89	0.50	10.94	9.93
	Fair	**6.97**	4.69	0.53	12.19	17.04
	Full	**8.70**	6.07	0.79	15.55	22.21
	Overlap	**15.23**	3.47	0.64	**19.35**	20.96
0.7, 4	Forgiving	5.16	4.64	0.49	10.29	9.97
	Fair	7.69	3.70	0.50	11.89	17.40
	Full	9.69	4.77	0.75	**15.21**	22.57
	Overlap	17.11	2.80	0.56	20.48	22.33

Experiments and results

Results on AliMeeting dataset

m, λ	Setup	Miss (↓)	FA (↓)	Conf. (↓)	DER (↓)	JER (↓)
baseline	Forgiving	2.52	2.53	0.37	5.42	7.04
	Fair	3.46	1.90	0.40	5.75	10.29
	Full	7.25	4.52	0.70	12.47	19.89
	Overlap	14.98	2.00	0.47	17.46	20.25
0.3, 4	Forgiving	2.49	2.23	0.37	5.09	6.87
	Fair	3.49	1.65	0.38	5.52	10.20
	Full	7.44	4.40	0.67	12.51	20.24
	Overlap	15.80	1.93	0.40	18.14	21.00
0.5, 4	Forgiving	**2.48**	**2.06**	**0.29**	**4.82**	**6.37**
	Fair	**3.32**	**1.47**	**0.30**	**5.10**	**9.26**
	Full	**7.03**	**4.32**	**0.57**	**11.92**	**19.06**
	Overlap	**14.60**	**1.83**	**0.35**	**16.77**	**19.50**

Results on CHiME-6 dataset

m, λ	Setup	Miss (↓)	FA (↓)	Conf. (↓)	DER (↓)	JER (↓)
baseline	Forgiving	**10.73**	61.51	**0.70**	79.20	**42.08**
	Fair	**14.72**	38.46	**7.46**	60.65	**49.28**
	Full	**17.48**	35.62	**8.64**	61.74	**52.89**
	Overlap	29.84	6.74	6.56	43.14	**51.68**
0.3, 4	Forgiving	16.57	**22.00**	8.58	**47.14**	45.81
	Fair	22.44	**12.27**	8.36	**43.07**	53.64
	Full	26.56	**12.25**	9.16	**47.96**	57.84
	Overlap	35.98	**3.26**	**6.14**	45.39	56.80
0.5, 4	Forgiving	16.36	38.68	8.58	63.61	47.34
	Fair	19.86	19.48	8.71	48.05	53.60
	Full	23.53	20.47	9.57	53.57	57.38
	Overlap	**26.54**	7.28	7.72	**41.55**	52.05

Heavy-tailed separation model

- Gaussian variance modeling assumes the Gaussianity of speech signal.
- However speech signal shows more **heavy-tailedness**.
- We consider a heavy-tailed extension to the variance modeling in the **separation model**!

Under the **Gaussian Scalar Mixture (GSM)** model formulation:

$$s_{nft}|r_{ft} \sim \mathcal{N}_{\mathbb{C}}(0, r_{ft}\lambda_{nft})$$

with an random impulsive variable r.
Marginalizing r with different prior $p(r)$ leads to different heavy-tailed model families, e.g.,

$$\begin{aligned} \mathbf{x}_{ft} &\sim \mathcal{GG}^{\beta}_{\mathbb{C}}\left(0, \mathbf{Y}_{ft}\right) \quad (\text{Leptokurtic}), \\ \mathbf{x}_{ft} &\sim \mathcal{T}^{\nu}_{\mathbb{C}}\left(0, \mathbf{Y}_{ft}\right) \quad (\text{Student's t}), \end{aligned}$$

Combined with the neural FCASA, we call this **Heavy-tailed FCASA** [Mao et al., 2026b][1]

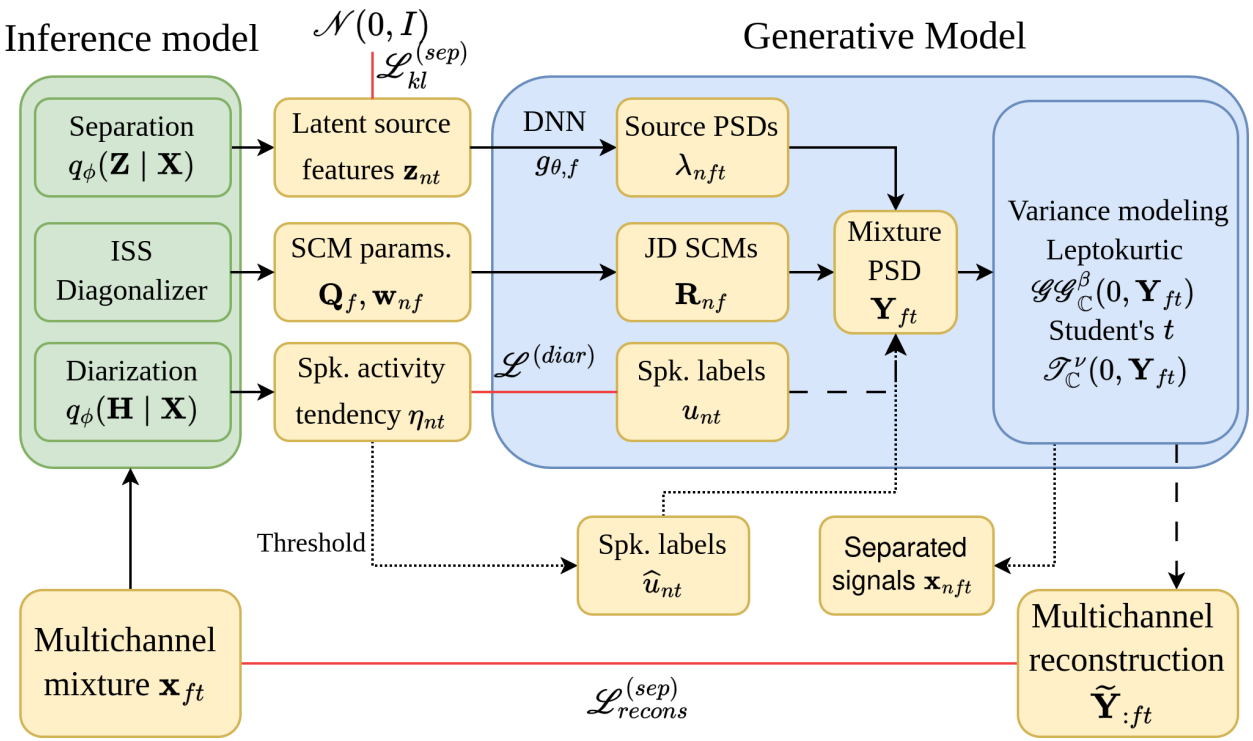
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1. Neural Multichannel Distant Speaker Diarization with Heavy-tailed Source Separation Model ↩

VI in Heavy-tailed FCASA

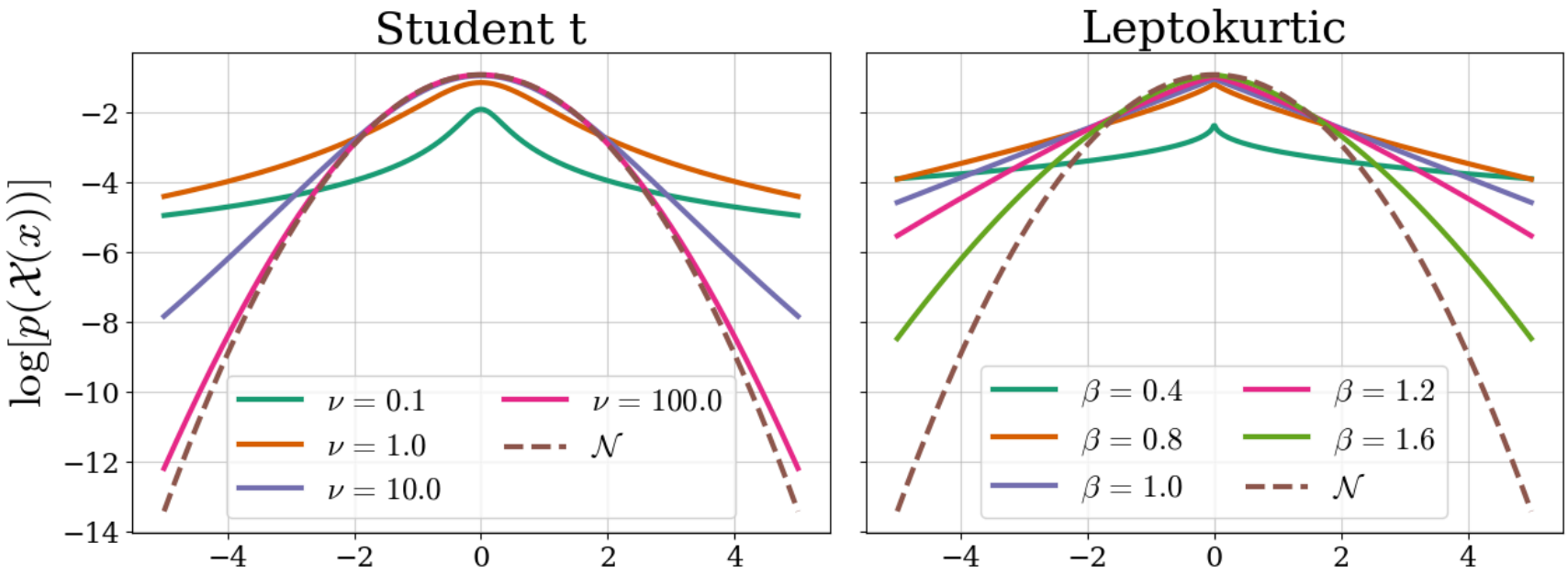
Inference model: same

$\mathcal{L}_{recons}^{(sep)}$ becomes to the heavy-tailed version.



Shape parameters of the heavy-tailedness

The Leptokurtic Generalized Gaussian distribution and the Student's t-distribution have **shape parameters** $\nu \in (0, +\infty)$ and $\beta \in [0, 2)$, respectively. The shape parameter controls the degree of heavy-tailedness.



Experiments and results

Choose different shape parameters for the heavy-tailed models and verify their effectiveness on the AMI dataset.

- Consistently better performance than the baseline (for certain shape parameters).
- Efficacy on AliMeeting and CHiME-6 datasets.

Results on AMI dataset

Model	Shape param.	Setup	Miss (↓)	FA (↓)	Conf. (↓)	DER (↓)	JER (↓)
Gaussian (baseline)	-	Forgiving	5.59	8.16	0.73	14.48	13.50
		Fair	8.61	6.33	0.79	15.73	23.16
		Full	10.77	6.81	1.16	18.73	27.65
		Overlap	19.40	3.80	0.90	24.11	26.53
Student's t	0.1	Forgiving	5.04	**5.59**	**0.61**	**11.24**	**10.78**
		Fair	7.49	**4.42**	**0.62**	**12.53**	**18.38**
		Full	9.23	5.86	**0.92**	**16.01**	**23.45**
		Overlap	16.18	**3.18**	**0.67**	20.03	21.99
	1	Forgiving	**4.30**	6.44	0.67	11.41	11.04
		Fair	**6.92**	5.02	0.69	12.62	18.80
		Full	**8.55**	6.52	0.98	16.05	23.82
		Overlap	**15.57**	3.44	0.70	**19.71**	**21.92**
	10	Forgiving	5.45	7.58	0.78	13.80	12.17
		Fair	8.55	6.17	0.80	15.52	20.98
		Full	10.93	5.99	1.07	17.99	25.95
		Overlap	19.76	3.71	0.74	24.21	26.01
	100	Forgiving	5.57	6.78	0.72	13.08	12.48
		Fair	8.93	5.39	0.80	15.12	22.09
		Full	11.25	5.96	1.09	18.30	26.82
		Overlap	20.12	3.55	0.78	24.44	26.72

Experiments and results

Results on AliMeeting dataset

Model	Shape param.	Setup	Miss (↓)	FA (↓)	Conf. (↓)	DER (↓)	JER (↓)
Gaussian baseline	-	Forgiving	2.52	2.53	0.37	5.42	7.04
		Fair	3.46	1.90	0.40	5.75	10.29
		Full	7.25	4.52	0.70	12.47	19.89
		Overlap	14.98	2.00	0.47	17.46	20.25
Student's t	0.1	Forgiving	2.28	2.35	**0.22**	4.85	5.97
		Fair	2.93	1.76	**0.22**	4.91	**8.46**
		Full	**6.38**	4.60	**0.49**	11.47	**18.11**
		Overlap	**13.42**	2.01	0.35	**15.78**	**18.28**
	1	Forgiving	**1.87**	**2.16**	0.28	**4.32**	**5.88**
		Fair	**2.62**	**1.59**	0.28	**4.49**	8.47
		Full	6.41	**4.09**	0.52	**11.03**	18.18
		Overlap	14.17	**1.76**	**0.33**	16.26	18.88

Results on CHiME-6 dataset

Model	Shape param.	Setup	Miss (↓)	FA (↓)	Conf. (↓)	DER (↓)	JER (↓)
Gaussian baseline	-	Forgiving	10.73	61.51	6.96	79.20	42.08
		Fair	**14.72**	38.46	7.46	60.65	49.28
		Full	**17.48**	35.62	8.64	61.74	52.89
		Overlap	29.84	6.74	6.56	43.14	51.68
Student's t	0.1	Forgiving	**9.16**	30.47	**5.75**	45.37	**32.40**
		Fair	14.83	19.30	**6.22**	40.35	**41.67**
		Full	17.85	29.01	**7.23**	54.10	**46.61**
		Overlap	**27.15**	10.95	6.14	44.25	**46.60**
	1	Forgiving	15.89	**19.20**	6.58	**41.67**	38.41
		Fair	20.52	**12.54**	6.80	**39.86**	46.97
		Full	23.72	**15.54**	7.57	**46.84**	51.44
		Overlap	32.61	**3.66**	**6.08**	**42.35**	51.89

Joint Heavy-tailed and Beta FCASA

We combine both improvements together as they are independent modifications on the neural FCASA.

- Choose $m = 0.5, \lambda = 4.0$ for the Beta FCASA and $\nu = 1$ for the Heavy-tailed FCASA with Student's t-distribution.
- Heavy-tailed FCASA and Beta FCASA model are compatible, and the joint FCASA model is better than individual improvements on AMI dataset.
- On AliMeeting and CHiME-6 datasets, the joint FCASA model consistently outperforms the baseline.

Model	Setup	Miss (↓)	FA (↓)	Conf. (↓)	DER (↓)	JER (↓)
baseline	Forgiving	5.59	8.16	0.73	14.48	13.50
	Fair	8.61	6.33	0.79	15.73	23.16
	Full	10.77	6.81	1.16	18.73	27.65
	Overlap	19.40	3.80	0.90	24.11	26.53
Student's t $\nu = 1$	Forgiving	4.30	6.44	0.67	11.41	11.04
	Fair	6.92	5.02	0.69	12.62	18.80
	Full	8.55	6.52	0.98	16.05	23.82
	Overlap	15.57	3.44	0.70	19.71	21.92
Beta $m = 0.5, \lambda = 4.0$	Forgiving	4.55	5.89	0.50	10.94	9.93
	Fair	6.97	4.69	0.53	12.19	17.04
	Full	8.70	6.07	0.79	15.55	22.21
	Overlap	15.23	3.47	0.64	19.35	20.96
joint	Forgiving	**4.29**	**4.81**	**0.38**	**9.49**	**8.75**
	Fair	**6.19**	**3.61**	**0.39**	**10.18**	**14.09**
	Full	**7.87**	**5.21**	**0.60**	**13.68**	**19.50**
	Overlap	**13.65**	**2.92**	**0.45**	**17.02**	**18.30**

Conclusions

Conclusions

In the joint modeling of separation and diarization:

- A heavy-tailed separation model improves the diarization performance.
- A Beta speaker prior model improves the diarization performance.
- Joint improvements are compatible and usually better than individual ones.

Limitations

- Heavy-tailed FCASA is sensitive to hyperparameter tuning.
- Optimal combination of joint FCASA hyperparameters are not well searched.
- Threshold selection for the diarization model is not investigated.
- No evaluation on separation metrics.

Open-source link

alephpi/neural-fcasa



Future Work

- Address aforementioned limitations.
- Incorporate geometric configuration of the microphone array.
- Incorporate more complex conversational dynamics.
- Incorporate speaker localization task.

END

Regularization of speaker embedding

Hierarchical latent variable model

We model the speaker embedding as generated from a hierarchical latent variable model.

$$\mathbf{y}_i \sim p(\mathbf{y}) = \mathcal{N}(\mathbf{y}; \mathbf{0}, \mathbf{I}) \tag{1}$$

$$\mathbf{z}_i|\mathbf{y}_i = \mathbf{y} \sim p(\mathbf{z}|\mathbf{y}) = \mathcal{N}(\mathbf{z}; f_{\boldsymbol{\theta}}(\mathbf{y}), \mathbf{I}), \tag{2}$$

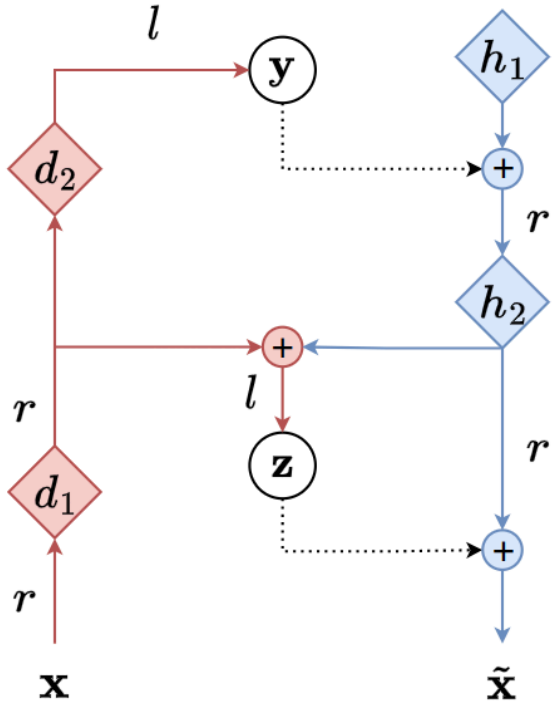
$$\mathbf{x}_i|\mathbf{z}_i = \mathbf{z} \sim p(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{x}; g_{\boldsymbol{\theta}}(\mathbf{z}), \mathbf{I}). \tag{3}$$

Inference by an HVAE

$$q_\phi(\mathbf{y}, \mathbf{z}|\mathbf{x}) = q_\phi(\mathbf{y}|\mathbf{x}) q_\phi(\mathbf{z}|\mathbf{y}, \mathbf{x}), \tag{4}$$

$$q_\phi(\mathbf{y}|\mathbf{x}) = \mathcal{N}(\mathbf{y}; \boldsymbol{\mu}_{\mathbf{y},\phi}(\mathbf{x}), \boldsymbol{\Sigma}_{\mathbf{y},\phi}(\mathbf{x})), \tag{5}$$

$$q_\phi(\mathbf{z}|\mathbf{y}, \mathbf{x}) = \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}_{\mathbf{z},\phi}(\mathbf{y}, \mathbf{x}), \boldsymbol{\Sigma}_{\mathbf{z},\phi}(\mathbf{y}, \mathbf{x})). \tag{6}$$



VI in HVAE

The ELBO function can be written as

$$\mathcal{L}(\mathbf{x};\theta,\phi) = \mathbb{E}_{q_\phi(\mathbf{y}|\mathbf{x})q_\phi(\mathbf{z}|\mathbf{y},\mathbf{x})}\left[\log p(\mathbf{x}|\mathbf{z})\right] - \mathrm{KL}(q(\mathbf{y},\mathbf{z}|\mathbf{x})\|p(\mathbf{y},\mathbf{z})) \tag{7}$$

where the first term is the reconstruction loss and the second term can be further expanded as

$$\mathrm{KL}(q(\mathbf{y},\mathbf{z}|\mathbf{x})\|p(\mathbf{y},\mathbf{z})) = \mathrm{KL}(q(\mathbf{y}|\mathbf{x})\|p(\mathbf{y})) + \mathbb{E}_{q(\mathbf{y}|\mathbf{x})}[\mathrm{KL}(q(\mathbf{z}|\mathbf{y},\mathbf{x})\|p(\mathbf{z}|\mathbf{y}))]. \tag{8}$$